Freshwater Biology

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Minimal Introgression but Restricted Gene Flow: How Stocking and Dams Influence Wild Brook Trout (*Salvelinus fontinalis*) Genetics and Morphology

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ABSTRACT

1. Anthropogenic activities are increasingly influencing eco-evolutionary processes across spatial, temporal, and biological scales. Freshwater ecosystems, harbouring species that are important for subsistence, recreational, and commercial fisheries, are particularly vulnerable. In North America, brook trout (*Salvelinus fontinalis*) populations have experienced widespread declines due to factors including habitat fragmentation and angling. To increase population densities and meet fishing demands, thousands of captive-bred individuals are released annually. Stocking, however, could lead to introgression of maladaptive genes into the wild population and, in combination with the presence of dams restricting gene flow, may reduce their evolutionary potential.
2. Here, we sampled 495 fish from 14 inland and two coastal sites in the state of Rhode Island, USA and from two hatchery strains. We used quaddRAD to develop single nucleotide polymorphism (SNP) markers and combined genetic and geometric-morphometric analyses to evaluate intraspecific variation, the consequences of stocking, and test hypotheses, in a riverscape framework, on how dams influence gene flow.
3. We found moderate genetic structure within streams with low levels of genetic diversity and effective population sizes but higher than the two hatchery strains. Importantly, we found no significant introgression from captive-bred individuals. However, we did detect isolation-by-distance and -by-resistance (due to dams) with evidence for upstream and downstream gene flow occurring mainly within catchments. Phenotypic differentiation also was apparent both between wild and hatchery brook trout and even among streams, correlating with genetic distances.
4. Our findings highlight the importance of fine-scale analyses in evaluating anthropogenic impacts and uncovering hidden intraspecific variation. Conservation efforts for wild brook trout populations, which harbour distinct genetic and phenotypic variation, will require careful attention, and our results will help inform management strategies across the species' range.

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1 | Introduction

Ecological and evolutionary processes influence the amount and structuring of genetic and phenotypic variability in natural populations and, while the determinants of such variation are part of species' evolutionary history, recent human activities have been shown to induce rapid phenotypic and genetic changes leading to major threats for biodiversity (Ceballos et al. 2020; Pelletier and Coltman 2018; Tilman et al. 2017). The intensity of anthropogenic activities on ecosystems through, for example, urbanisation, habitat alteration and the introduction of non-native species, may vary and the impact on natural populations could be particularly strong (Pelletier and Coltman 2018). For example, in fisheries, the selection pressure imposed by intense harvest may cause shifts in growth rates and reproductive timing in many commercially important species, reducing yields and species' capacity to recover from overfishing (Heino et al. 2015). In addition to demographic changes, human-induced microevolutionary changes also occur in populations, although species' responses are often attributed to phenotypic plasticity (Merilä and Hendry 2014; Pelletier and Coltman 2018). Nevertheless, intraspecific variation, which plays an important role in eco-evolutionary processes, has been typically overlooked in monitoring programs (Des Roches et al. 2018; Mimura et al. 2017). Understanding of how genetic and phenotypic variability is distributed in space and the impact of recent human activities is fundamental in our assessment of a species' capacity to adapt to novel anthropogenic or environmental pressure, but also to define conservation units (Hohenlohe et al. 2020).

Habitat fragmentation in freshwater systems can occur as a natural process through, for example, watersheds, which act as barriers to gene flow and influence connectivity, shaping the population genetic structure of fishes within and between catchments (e.g., Bruce et al. 2019). However, anthropogenic activities, such as the construction of dams, can exacerbate habitat fragmentation by disrupting natural migratory processes and gene flow among populations (Kelson et al. 2015; Waples et al. 2008), and can have significant ecological effects, for example, through changes of thermal regimes (Olden and Naiman 2010). The obstruction of gene flow within catchments, rivers and hydrological networks can lead to population isolation, reducing genetic diversity and increasing the risk of inbreeding, which may impede the adaptive response to novel selective pressures (Bijlsma and Loeschcke 2012) and ultimately result in population decline and local extinction (Frankham et al. 2017).

Stocking of freshwater fish species is a common practice in North America to support the recreational activity of millions of anglers and/or as restorative supplementation to bolster the abundance and/or genetic diversity of wild populations. In the case of supplementation, the goal is to increase the number of adult fish that spawn in the wild in subsequent years (Claussen and Philipp 2023). However, due to the selective process in the hatcheries favouring specific traits, captive-bred stocks often have reduced genetic diversity, a non-native origin and likely have lower fitness than their wild counterparts (Araki et al. 2008; Claussen and Philipp 2023). If captive-bred stocks interbreed with wild populations, maladaptive genes could be transferred to wild populations, which might have serious negative outcomes for already depauperate populations (Araki and Schmid 2010; Claussen and Philipp 2023; Fraser 2008).

The brook trout (*Salvelinus fontinalis*) is an ideal study system to address how anthropogenic activities have influenced connectivity among populations and the resulting consequences at both the genomic and phenotypic levels. Wild brook trout populations are thought to have declined across their native range in Northeastern North America primarily due to habitat fragmentation, caused by dams and other physical barriers, which have hindered the natural flow of waterways (EBTJV 2006; Hudy et al. 2008). The construction of dams and other anthropogenic causes has also negatively impacted brook trout by increasing the water temperature and decreasing the water quality of their habitat (Bassar et al. 2016; EBTJV 2006; Hudy et al. 2008; Turgeon et al. 2019). Brook trout also are a popular freshwater fish for anglers and annually, thousands of hatchery-raised individuals are released into the wild. The genetic impact of stocking hatchery-reared brook trout into wild brook trout habitat has been extensively studied, with some studies finding no or little evidence for introgression (Annett et al. 2012; Kelson et al. 2015; Létourneau et al. 2018; White et al. 2018), while others have found fairly extensive introgression between stocked and wild brook trout (Beer et al. 2019; Bruce, Kutsumi, et al. 2020; Lamaze et al. 2012; Valiquette et al. 2014). This indicates that the effects of stocking can depend on the details of the supplementation program employed in each region. In addition, brook trout exhibit various life history strategies throughout their widespread native range (e.g., anadromous and fluvial) and populations might respond differently to anthropogenic pressure. Fewer studies have simultaneously examined the phenotypic diversity within and divergence between wild and captive-bred brook trout (e.g., Gossieaux et al. 2020), which also might determine the impact of stocking. Lastly, because brook trout prefer clear, cold-water streams, they can also indicate the relative health of their habitat and water quality (EBTJV 2006). Thus, studying and conserving brook trout can have additional benefits for other cold-water fish species and other flora and fauna associated with their habitat (Beer et al. 2019).

In Rhode Island (RI), USA, brook trout are managed by the Rhode Island Department of Environmental Management (RIDEM) Division of Fish and Wildlife (DFW), which lists brook trout as a Species of Greatest Conservation Need (RI SGCN 2015). Annually, the DFW releases thousands of hatchery-reared brook trout from two main hatchery strains: Lafayette Rome (LFR) and Assinica (LFA) to meet the fishing demands of anglers. These stocks are released as put-and-take, and they are assumed to be less fit than wild brook trout. DFW reports from their stream electrofishing surveys (conducted between June and August) indicate that few to no stocked trout are captured, despite the stocking of hatchery-reared trout between March and May (Alan Libby, personal observations). However, fine-scale genetic analyses to assess the status of wild brook trout populations have yet to be conducted. Rhode Island also is an ideal region to study the impact of habitat fragmentation because the state has a long history of dam construction that dates back to the European colonisation. In 2019, RIDEM listed 669 state regulated dams in RI with an average year built of 1896 (±177 SD; RIDEM 2019).

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The main objective of our study was to characterise the extent of genomic and phenotypic divergence between wild and hatchery-reared brook trout and to elucidate the effects of stocking and habitat fragmentation in RI streams. More specifically, the goals of this study were to determine: (1) if introgression has occurred between wild and captive-bred brook trout; (2) the impact of dams on brook trout gene flow; and (3) if the genetic divergence of wild brook trout is reflected in their morphology. We do not expect to find introgression between hatchery stock and wild brook trout, despite extensive stocking, because of the suspected low survival rates of hatchery fish. We do expect wild brook trout to display population structure and genetic differentiation that correlates with the presence of dams as main barriers to fish movement. Because of asymmetrical, downstream-biased dispersal, we expect a downstream increase in genetic diversity (DIGD) (Paz-Vinas et al. 2015). Although captive stocks are usually physically much larger than wild brook trout, we also expect to find morphological differences between the wild and hatchery fish independent of body size. To address our goals, we developed single nucleotide polymorphism (SNP) markers using a quaddRAD (Franchini et al. 2017) approach on 495 brook trout from 14 inland streams and two coastal sites in RI as well as from two hatchery strains that have been released in RI annually over the past decade. We also characterised the morphology of a subset of wild brook trout (from both inland and coastal sites) and the two hatchery strains using geometric-morphometric analyses of photographs. Information about the genetic connectivity of the river systems in RI will provide baseline information that could be used to determine the efficacy of efforts to restore the connectivity of rivers and identify areas where wild brook trout may be suffering the negative effects of inbreeding due to isolation from other populations.

2 | Methods

2.1 | Study Sites and Sampling

We surveyed 16 sites in RI between June and October 2017 (Table 1; Figure 1). We included sites where only wild brook trout reside (non-stocked sites; $n = 6$), as well as sites where stocking with hatchery-reared brook trout occurred (stocked sites; $n = 8$) within the past 10 years. We also included two coastal sites (SBD and SHB; Figure 1) to compare genetic and phenotypic diversity and divergence. Both coastal sites have open connectivity with their respective estuaries and may include anadromous brook trout, though their status remains unknown (Dauwalter et al. 2014). At each site, sampling was distributed across a minimum of 100 m to reduce the likelihood of oversampling family groups, and when possible, multiple river segments were included. We caught 434 wild brook trout via backpack electrofishing and fish were immediately placed in buckets with freshwater and aerators. In addition, for the comparative analyses, we collected 61 samples from two hatchery strains maintained at the Lafayette Fish Hatchery in RI; the Assinica strain (LFA), originating from Lake Assinica in Quebec Canada, and the Rome strain (LFR) produced by Rome Fish Hatchery in New York state. From each fish (wild caught and from the hatchery), we collected total length (±1 mm), a tissue sample from the pelvic or caudal fins (stored in 95% ethanol) and two photographs while each fish was on a measuring board for the geometric-morphometric analyses (see details below). After sampling, all fish were returned to the water near the location of capture and visually monitored to ensure they recovered (Institutional Animal Care and Use Committee protocol: AN1819-001).

2.2 | Laboratory Work

Genomic DNA was extracted from the fin clips using either the NucleoMag 96 Tissue kit (Macherey Nagel, Bethlehem, PA) or the DNeasy Blood & Tissue Kits (Qiagen, Valencia, USA) following the manufacturer's instructions. Next-generation library preparation for SNP discovery was conducted using the quaddRAD protocol (Franchini et al. 2017) with modifications. Briefly, genomic DNA was quantified using fluorescent kits Quant-it dsDNA Assay kits (Invitrogen) or Accublue dsDNA kits (Biotium) on a Synergy H1 plate reader (Biotek). A minimum of 300 ng of DNA was digested using the *EcoRI-HF* and *SphI-HF* restriction enzymes (New England Biolabs) and subsequently, fragments were then ligated with uniquely barcoded adaptor pairs (I5 and I7, Illumina). We performed individual polymerase chain reactions (PCRs) using the Kapa Library Amplification Master Mix (Roche) with unique primer pairs (i5- and i7-, Illumina) for 12–14 PCR cycles. PCR products were quantified using Quant-it dsDNA Assay kits or Accublue dsDNA kits and pooled equimolarly in two library pools (library-A: 280 samples, library-B: 279 samples). We included replicate samples across both libraries ($n = 128$) to check genotype error rates. Libraries were size-selected (370–605 base pairs) using BluePippin (Sage) and ran on a BioAnalyzer (Applied Biosystems) to verify the distribution of fragments. We sent the two library pools (559 individually-barcoded libraries in total) to Genohub-Admera Health (New Jersey, USA) for 150 base pair paired-end sequencing using a HiSeq 4000 (Illumina) in two lanes.

2.3 | Sequencing Data Quality and Variant Calling

We first used the *clone filter* module in STACKS 2.5 (Rochette et al. 2019) to remove PCR duplicates (4% of total reads) followed by the *process radtags* module to demultiplex sequenced data and filter out poor-quality reads for each individual based on barcode information. Options to clean data (*–c*), rescue barcodes (*–r*) and discard reads with low-quality scores (*–q*) were applied with default values. Sequence quality assessment was performed using the software FastQC (Andrews 2010). Next, we employed the *denovo_map* pipeline in STACKS. Initially, we tested the assembly parameters by running the pipeline on a subset of samples ($n = 128$). Values for the three main parameters *–m* (minimum number of identical raw reads to create a stack), *–M* (number of mismatches allowed between two alleles of a heterozygote) and *–n* (the number of mismatches between any two alleles) were chosen following the optimization procedure described by Rochette and Catchen (2017). Briefly, *–m* was set to three (default), and increasing values for *–M* and *–n* were tested. The combination of these parameters for which the number of polymorphic loci present in at least 80% ($R > 0.8$) of the individuals reached a plateau was defined as the optimal parameter set. We also used the replicated samples dataset ($n = 128$) to estimate genotypic errors by performing a relatedness analysis,

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TABLE 1 | Sampling sites and summary statistics.

Site	Code	Longitude	Latitude	STK	Dam Position	N_{GEN}	N_{PHE}	H_O	H_E	P_I	F_{IS}	N_E
Acid Factory Brook	AFP	−71.721953	41.633769	No	Above dam	20	—	0.273	0.264	0.271	−0.001	80.9 (30.3–infinite)
Baker Brook	BAK	−71.693075	41.542195	Yes	Below dam	30	—	0.292	0.292	0.297	0.022	240.6 (123.1–2475)
Breakheart Brook above Breakheart Pond	BHU	−71.697832	41.604672	No	Above dam	25	—	0.255	0.250	0.255	0.005	68.4 (35.5–341)
Beaver River at Hillsdale Road	BRH	−71.640094	41.522293	No	Below dam	27	—	0.262	0.256	0.261	0.003	111.5 (54.8–1491.9)
Beaver River at Old Mountain Road	BRO	−71.640852	41.538449	No	Above dam	25	—	0.262	0.254	0.260	0.002	37.3 (20.1–117.9)
Beaver River at Punchbowl Trail	BRP	−71.64156	41.512152	Yes	Below dam	29	31	0.277	0.270	0.275	0.000	76.2 (40.6–320)
Brushy Brook	BRU	−71.736417	41.528783	Yes	Below dam	27	28	0.275	0.278	0.284	0.048	266.9 (99.5–infinite)
Falls River	FRC	−71.735044	41.588722	Yes	Below dam	29	—	0.300	0.296	0.301	0.010	175.7 (77.8–infinite)
Meadow Brook Carolina Management Area	MBC	−71.689147	41.457545	Yes	Below dam	29	—	0.284	0.278	0.284	0.004	65.3 (37.8–179.7)
Meadow Brook at Pine Hill Road	MBP	−71.688469	41.471734	Yes	Below dam	16	—	0.275	0.277	0.286	0.036	49.1 (20.2–infinite)
Smelt/Brown Brook below Rum Pond	SBD	−71.512712	41.414437	No	Coastal	15	24	0.245	0.223	0.231	−0.028	22.3 (13.2–50.3)
Shelter Harbour Brook	SHB	−71.738029	41.341279	No	Coastal	30	25	0.192	0.185	0.189	0.000	63.3 (37.9–154.2)
Unnamed tributary to Wood River below Alton Dam	UTA	−71.723342	41.437903	Yes	Below dam	17	29	0.298	0.285	0.294	−0.002	509.2 (61.3–infinite)
Unnamed tributary to Breakheart Pond (above dam)	UTB	−71.699931	41.595471	No	Above dam	27	28	0.249	0.242	0.247	0.003	65.5 (42.8–126.2)
Unnamed tributary to Breakheart Pond (below dam)	UTB2	−71.700343	41.59587	No	Below dam	25	—	0.254	0.246	0.251	−0.003	44 (25.9–108.9)

(Continues)

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TABLE 1 | (Continued)

Site	Code	Longitude	Latitude	STK	Dam Position	N_{GEN}	N_{PHE}	H_O	H_E	P_I	F_{IS}	N_E
Wood River upstream of Frying Pan Pond	WR	−71.714549	41.565066	Yes	Above dam	21	24	0.301	0.294	0.301	0.006	278.4 (93.4–infinite)
Lafayette Hatchery Assinica strain	LFA	—	—	—	—	23	30	0.232	0.219	0.224	−0.014	17.8 (13.3–25)
Lafayette Hatchery Rome strain	LFR	—	—	—	—	29	31	0.271	0.272	0.277	0.023	33.6 (21.7–62.1)

Abbreviations: STK, whether the site has been stocked with hatchery fishing in recent years; Dam position, the location of the site in relation to the dam (above or below); number of genotyped individuals post-filtering, N_{GEN}; number of individuals included in morphological analyses, N_{PHE}; mean observed heterozygosity, H_O; mean expected heterozygosity, H_E; nucleotide diversity, P_I; mean individual inbreeding coefficient, F_{IS}; and effective population size, N_E with 95% CIs.

Identity-by-State (IBS), on SNP genotypes in the R package *SNPrelate* (Zheng et al. 2012).

2.4 | Filtering and Final Datasets

First, we removed poor quality samples with less than 200,000 reads and the replicated samples. Following our initial test, we selected the optimal value for the assembly parameters ($M = n = 3$, see Figure S1 and results section) and ran the *denovo_map* pipeline on the remaining samples with a minimum minor allele frequency (*min_maf*) of 0.05 and $R > 0.8$ (minimum percentage of individuals across populations required to process a locus). Then, we calculated the number of missing SNPs per individual using VCFtools v0.1.15 (Danecek et al. 2011) and filtered out those individuals with more than 25% of missing data. We repeated the *denovo_map* pipeline on the final dataset, which included 444 individuals.

To focus on neutral evolutionary processes (i.e., genetic drift, gene flow) in our analysis of genetic patterns, we used the R package *pcadapt* (Luu et al. 2017) to identify and remove SNPs potentially under selection (alpha threshold set to 0.1). We also used the *dDocent* pipeline (Puritz et al. 2014) to remove SNPs that were out of Hardy–Weinberg equilibrium (minimum cutoff *p*-value: −*h* 0.01). Loci identified from either analysis ($n = 1070$) were combined into a blacklist and used in the *populations* module in STACKS to output various datasets for subsequent analyses. The first dataset (*BT_All*) contained all post-filtering individuals ($n = 444$) and all SNPs for population genetic diversity estimates (see below). For finer scale analyses focusing on inland RI sites, we created a second dataset (*BT_Inland*; $n = 347$) excluding individuals from the two hatchery strains (LFR and LFA) and the two coastal RI sites (SHB and SBD). To meet the assumptions of linkage equilibrium in clustering methods (see below) we used the *write_single_SNP* option in the *populations* module in STACKS to output the respective datasets. The complete dataset (*BT_All*) is available on figshare (see also Data Accessibility Statement).

2.5 | Genetic Diversity, Population Differentiation and Structure

Population genetic diversity indices: observed (H_O) and expected heterozygosity (H_E), inbreeding coefficient (F_{IS}) and nucleotide diversity (P_I) were calculated with the *populations* module (*fstats* option) in STACKS using the full dataset (*BT_All*, with all the SNPs). We also calculated the observed heterozygosity per individual using VCFtools and compared genetic diversity between groups (hatchery vs. wild, vs. stocked, vs. non-stocked and vs. coastal sites) using non-parametric tests in R (R Core Team 2023). Pairwise population differentiation (F_{ST}) was calculated in the R package *DiveRsity* (Keenan et al. 2013) and we used the pairwise F_{ST} matrix to visualise population relationships with a neighbour-joining tree using the R package *ape* (Paradis et al. 2004).

We used multiple complementary methods to quantify and describe the extent of genetic population structure across our sampling sites. First, we used a Bayesian analysis implemented in

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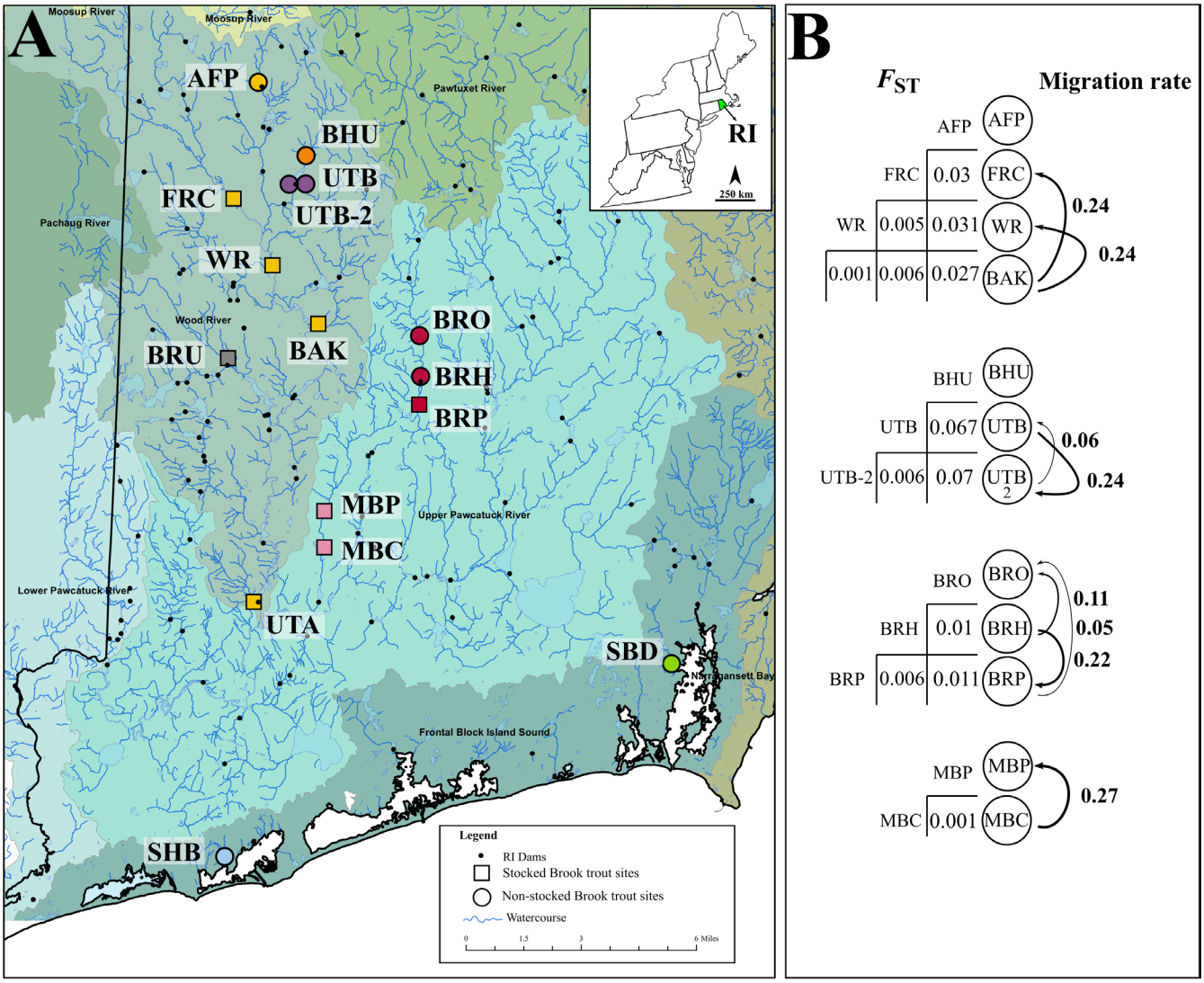

FIGURE 1 | Sampling sites and pairwise gene flow estimates. (A) Distribution of 16 sites in Rhode Island (RI) sampled for genetic analyses. Stocked and non-stocked sites are represented by a square and circle respectively and are colour-coded based on the fastSTRUCTURE results (see main text and Figure 2). Black dots represent dams. The inset map shows RI state (highlighted in green) in northeastern USA. (B) Pairwise genetic differentiation (F_{ST}, see also Table S3) and significant migration rates (the fraction of individuals that are migrants from another site, per generation) among sites used in the BayesAss analysis. Arrows show directionality in gene flow and the thickness of the line corresponds to the migration value. Catchment delineation (background shades of green) data are from USGS HUC12 Watershed Area Boundary Dataset through the Rhode Island GIS (RIGIS) and the watercourse data were derived from the HYDRO_hydrolines shapefiles via RIGIS. The dams (black dots) were derived from the Rhode Island Department of Environmental Management (RIDEM) as a DAM_layer of all registered dams in RIGIS.

fastSTRUCTURE (Raj et al. 2014) to detect the number of genetic clusters (K) without any prior information regarding population identity or geographic location. The upper K limit was set to the number of sampling sites ($K = 18$ for the full dataset and $K = 14$ for the inland RI subset), and the *chooseK.py* function was used to infer the most likely value of K. We used the R package *Pophelper* v1.2.0 (https://github.com/royfrancis/pophelper) to generate bar plots. Genetic structure also was assessed with a principal component analysis (PCA) using the function '*glPCA*' implemented in the R package *adegenet* 2.1.2 (Jombart and Ahmed 2011) on both filtered datasets (*BT_All* and *BT_Inland*). We also ran a Discriminant Analysis of Principal Components (DAPC, Jombart et al. 2010) using the 'dapc' function implemented in the R package *adegenet* 2.1.2. The '*find.clusters*' function was first used to determine the optimal K with the lowest Bayesian information criterion (BIC) and the '*optim.a.score*' function to choose the number of principal components (PCs) to retain to avoid model overfitting. For the DAPC analysis, we used the full dataset because the method does not require loci to be in linkage equilibrium. Finally, we used *fasta* sequences (consisted of 16,226 SNPs) of all 444 individuals to construct a NeighborNet (Bryant and Moulton 2004) network (i.e., split graphs) from uncorrected P-distances using the program SplitsTree V.4.14.6 (Huson 1998; Huson and Bryant 2006). This method is suitable for inferring evolutionary histories that are still subject to gene flow.

To investigate possible hybridization between captive-bred and wild-caught fish, we used the proportion of membership (q-value, range 0–1) to a given cluster from the fastSTRUCTURE

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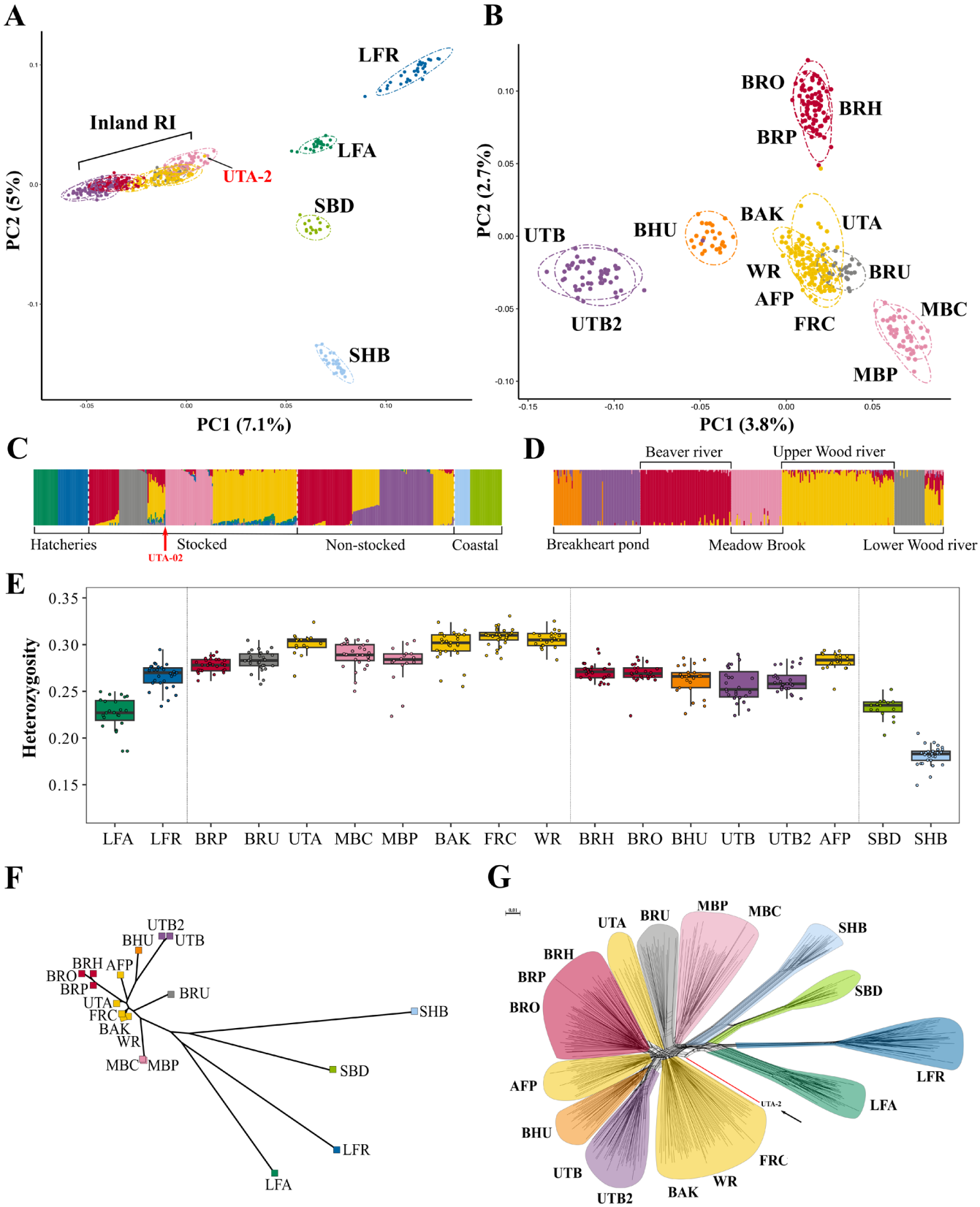

FIGURE 2 | Population genomic diversity and structure of wild and hatchery-reared brook trout in Rhode Island. (A) First two principal components summarising genetic variation based on 9770 SNPs for the full dataset (*BT_All*) and (B) for the inland dataset (*BT_Inland*). (C) Bayesian clustering plots from fastSTRUCTURE for the full dataset ($K=9$) and (D) for the inland RI ($K=6$). (E) Boxplots of heterozygosity (observed) across populations which are ordered as in panel C: Hatcheries, Stocked, Non-stocked, and Coastal. (F) neighbour-joining tree based on the pairwise F_{ST} distance matrix and (G) SplitsTree plot showing structuring of all sampled individuals. Colour scheme used throughout comes from combined fastSTRUCTURE results (at $K=9$ and $K=6$). Arrows point to the single individual (UTA-2) that showed evidence of captive-bred allele introgression.

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analysis. We first used the q-values from the optimal K, as inferred from the full fastSTRUCTURE run, and considered a threshold q-value of 0.2 to be suggestive of a hybrid individual (Vähä and Primmer 2005). To increase the accuracy of detecting true hybrids, we repeated the fastSTRUCTURE analysis independently for each stocked site or group of sites belonging to the same genomic cluster (BRP, BRU, MBP-MBC, BAK-FRC-WR and UTA) and the two hatchery strains (LFA and LFR). We set K to three (the two hatchery strains formed independent genetic clusters in the full analysis and because the wild sites, MBP and MBC, as well as BAK-FRC-WR formed a single genetic cluster we combined them for this analysis, see Results below) and ran the fastSTRUCTURE analysis three times. We combined runs and used a stringent threshold q-value of 0.1 to be suggestive of introgression (Vähä and Primmer 2005).

2.6 | Effective Population Size and Migration Rates

Effective population sizes (N_E) were estimated for each sampling site (wild and captive-bred) using the linkage disequilibrium model (LDNe) based on single moment data as implemented in NeEstimator v2 (Do et al. 2014). Mean N_E estimates were made using an allele frequency cutoff (*P-crit*) of 0.05, and 95% confidence intervals (CI) were established from jackknife cross-validations.

We used BayesAss v 3.1.1 (Wilson and Rannala 2003), which is modified to allow analysis of large SNP data sets (https://github.com/smussmann82/BayesAss3-SNPs), to estimate recent immigration among population groups. The program uses a Bayesian method with Markov chain Monte Carlo (MCMC) simulations to provide estimates of the mean and 95% CI of recent immigration, where migration (m) is the proportion of each population having migrant ancestry. Mixing parameter values for migration rates (deltaM), allele frequencies (deltaA), and inbreeding coefficients (deltaF) were adjusted to achieve recommended acceptance rates of 20%–40%. We performed five replicated runs of 6×10^6 MCMC iterations with 2×10^6 iterations discarded as burn-in and a sampling frequency of 1000. Each run was initialized with different starting-seed values to achieve consistency of mean parameter estimates between runs (Wilson and Rannala 2003). We performed fine-scale analysis by grouping sites belonging to the same catchment: upper Wood River (AFP, FRC, WR, BAK), Beaver River (BRH, BRP, BRH), Meadow Brook (MBC, MBP) and Breakheart Pond tributaries (BHU, UTB, UTB2). Convergence between runs was confirmed by comparing the posterior mean estimates for concordance, and significant migration rates were accepted when CIs were not overlapping zero. We also visualised the MCMC trace for each run with Tracer v.1.7 (Rambaut et al. 2018).

2.7 | Riverscape Genetics

We tested the potential influence of riverscape features (i.e., waterway distance and presence of dams) on population genetic diversity and gene flow using multiple approaches. First, we used linear regression to test the theoretical expectation that a downstream-biased, asymmetrical dispersal will generate a spatial pattern of a downstream increase in genetic diversity (DIGD) in river networks (Paz-Vinas et al. 2015). To do so, using ArcGIS v10 (ESRI 2011), we measured the waterway distance (W_{Dist}) of each inland site (i.e., excluding the two coastal sites) from the downstream-most sampling site at UTA (Figure 1). We obtained catchment (watershed area) information from the USGS HUC12 Watershed Area Boundary Dataset via the Rhode Island GIS (RIGIS) and watercourse centerline data and measurements were derived from the HYDRO_Hydrolines shapefile via RIGIS. Waterway distances were measured using the 'calculate geometry' function in ArcMap v10.4. The strength of the DIGD correlation was measured using Pearson's coefficient in R (R Core Team 2023). We also used the RIDEM_Dams layer of registered dams via RIGIS and counted the number of dams (D) between UTA and all other inland RI sites (Figure 1) to test for a correlation with genetic diversity. Second, we used a non-parametric test to check for a relationship between geographic connectivity and genetic diversity. To this end, we categorised each inland sampling site as above or below a dam to indicate the position of the site relative to the barrier.

Multiple regressions and Mantel tests were used to measure the effects of riverscape variables (i.e., waterway distance and number of dams) on genetic differentiation (F_{ST}) between sites. Waterway distance (W_{Dist}) and the number of dams (D) between all inland RI sites were calculated using ArcGIS v10 as above (Table S6, Table S7) and we generated pairwise distance matrices (i.e., W_{Dist} and D_{Dist}) in R. Then, we considered three models (M) to explain genetic differentiation (F_{ST}): M1 as a function of D_{Dist} between sites (isolation-by-resistance model), M2 as a function of W_{Dist} (isolation-by-distance model) and M3 testing the combined effect of waterway distance and presence of dams. Regression models were evaluated and ranked based on ΔAICc using the package *MuMIn* (Barton 2012). Mantel tests (Pearson's product–moment correlation with 9999 permutations) between the same variables in each model were performed with the *Vegan* package (Oksanen et al. 2025) in R (R Core Team 2023).

2.8 | Geometric-Morphometric Analyses

A subset of 251 fish from seven stream sites plus the two hatchery strains were analysed for body shape (Table 1). We used landmark-based geometric-morphometric methods, implemented in MorphoJ (Klingenberg 2011). We selected 17 landmarks (Figure S2) that characterise variation in trout body shape, fin positioning, head shape, plus eye size and position, and so are plausibly related to functional characteristics such as feeding, swimming speed and manoeuvrability (Bertrand et al. 2008; Imre et al. 2002; Rouleau et al. 2010). Similar sets of landmarks have been used for previous morphometric analyses of brook trout (Varian and Nichols 2010) and other salmonids (Arbour et al. 2011; Vehanen and Huusko 2011; e.g., Winans 1984). Fish were photographed live while on a measuring board. Two digital images of each fish's left side were taken and the XY coordinates of the landmarks were digitised from the images using the PointPicker plugin for ImageJ2 (Rueden et al. 2017; Thévenaz 2016).

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The XY coordinates were converted to shape variables using a generalised least-squares Procrustes superimposition, which calculates an average shape and aligns images to this average shape to remove variation in position and orientation from the coordinates (Dryden and Mardia 2016). Of 486 images initially screened, 23 were excluded because of distortion or poor focus and were confirmed as outliers by inspection of the cumulative distribution of the Procrustes distances of individual specimens from the average shape of the entire sample (Klingenberg and Monteiro 2005). We used Procrustes ANOVA to assess measurement error, measured as the difference between the two images of each fish, relative to biological sources of variation (Viscosi and Cardini 2011). The ANOVA sums of squares (as % of the total) indicated more variation between streams (32%) and individuals (53%) than images (14%), suggesting moderate measurement error (1.1%). To minimise any effect of measurement error, the two landmark configurations for each fish were averaged to make individuals the unit of study in further analyses. A pooled within-group regression also revealed a significant effect of body size on shape (approximate $r^2 = 0.058$; permutation test, $p < 0.001$). To remove effects of body size, the residuals from this regression were thus used in subsequent tests for differences in shape among groups (Klingenberg 2016).

Differences in body shape between groups were tested using canonical variate analysis (CVA) and differences between group means were assessed using permutation tests of Mahalanobis differences and Procrustes distances. For each pair of streams, discriminant function analysis (DFA) was used to explore if individuals could be correctly reassigned to their source group based on principal components of shape (Klingenberg and Monteiro 2005), and the reliability of group assignments was assessed by a jackknife (leave-one-out) cross-validation. Finally, we tested for an association between phenotypic (Procrustes distance) and genomic distance (F_{ST}) among sites with Mantel tests.

3 | Results

3.1 | Sequencing Output and Final SNP Data

We obtained a mean of 1,250,868 raw reads per sample across both sequencing lanes (559 samples, SD = 859,374.57). From the set of test runs using the replicate samples dataset ($n = 128$), we found that for M and n values greater than three (3), the number of loci plateaus and the distribution is mostly unaffected (Figure S1). Therefore, we applied the *denovo* assembly to the full dataset using $M = n = 3$. Relatedness analyses of the replicate samples estimated low genotypic error (< 1%). After removing replicates, samples with a low number of reads (< 200,000) and those with more than 25% of missing data, the final full dataset consisted of 444 samples genotyped at 160,832 loci with a mean coverage of 48.7×. After removing 1070 outlier loci and applying population filtering, 18,970 loci were kept with 16,226 variant sites (SNPs) in the full dataset (*BT_All*, keep all SNPs) and 8573 SNPs in the single-SNP-per-locus dataset. For the inland RI dataset, the final filtering produced 8097 SNPs (single-SNP-per-locus, see Table S1 for a summary).

3.2 | Genetic Diversity, Population Differentiation, and Structure

Genetic diversity (observed and expected heterozygosity and nucleotide diversity) varied among sites, with one of the coastal sites (SHB) having the lowest values across all indices (Table 1; Figure 2E; Table S2). Wild populations typically had higher levels of genetic diversity (i.e., heterozygosity) than the two hatchery strains (Figure S3) albeit non-significant (Wilcoxon rank sum exact test, $p > 0.21$). The two coastal sites (SHB and SBD) had significantly lower levels of genetic diversity than the inland sites (Figure 2E; Figure S3). Excluding the two coastal sites (also non-stocked), the stocked sites in RI had higher genetic diversity (observed and expected heterozygosity) compared to non-stocked sites (Figure S3: Wilcoxon rank sum exact tests $p < 0.05$). The inbreeding coefficient was generally low among all RI sites (Table 1).

Pairwise differentiation (F_{ST}) among sites ranged between 0.0014 and 0.378 (Table S3: mean = 0.14, SD = 0.10). The greatest F_{ST} values were between the coastal sites (SHB and SBD) and between the two hatcheries (LFA and LFR) and the other RI sites (Figure 2F; Table S2). The lowest values ($F_{ST} < 0.05$) occurred between paired sites within the same catchment (Figures 1 and 2F, Table S2), which also grouped closely in the neighbour-joining network (Figure 2F). Only two pairs (MBC vs. MBP and WR vs. BAK) did not differ significantly (Table S2).

To characterise genetic structure among sampling locations, we first employed a PCA using the full dataset (with a single-SNP-per-locus). The analysis showed five main groups (Figure 2A) with the first two axes explaining ~12% of the variation. The two hatchery strains (LFA and LFR) and the two coastal sites (SHB and SBD) formed independent groups. The remaining inland RI sites grouped together, but with subdivision along the first axis (PC1). Running the analysis on these inland sites (subset *BT_Inland*), the two PC axes explained ~7% of the variation (Figure 2B). Four subgroups were identified: subgroup 1 (UTB and UTB2), subgroup 2 (MBC and MBP), subgroup 3 (BRO, BRP and BRH) and subgroup 4 (AFP, FRC, WR, UTA and BAK) with BRU slightly differentiating from subgroup 4 and BHU on its own. The DAPC analyses showed similar results, with eight groups for the full dataset and four for the inland RI sites (Figure S5).

The Bayesian analysis in fastSTRUCTURE supported nine clusters ($K = 9$) for the full dataset (Figure 2C) and six clusters ($K = 6$) for the inland RI subset (Figure 2D). For the full dataset, the two hatchery strains (LFA and LFR) and the two coastal sites (SHB and SBD) formed their own clusters. Within RI, six additional clusters grouped sites belonging to the same catchment: Breakheart Pond tributaries, Beaver River, Meadow Brook, Upper Wood River tributaries and Wood River sites. For the second dataset (inland RI sites), there was further subdivision with BHU forming an independent cluster (Figure 2B), which also was separated at $K = 10$ in the full dataset. Similarly, the network analyses in SplitsTree supported the clustering analyses (Figure 2G). We found strong differentiation between the two hatcheries and the two coastal sites in relation to the inland RI streams, with sites

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belonging to the same catchment appearing more connected. Interestingly, one individual (UTA-2) was an outlier, appearing closer to the hatchery networks (Figure 2G), suggestive of possible introgression (see also below).

The proportion of membership (q-value) in a defined cluster (from the fastSTRUCTURE analyses) was used as a measurement of individual admixture (gene flow between captive-bred and wild fish). First, using results from $K = 9$ (full dataset analysis) and applying a threshold of $q > 0.2$, we found only one individual (UTA-2) with 27% combined membership to the two hatchery-strain clusters (23% to the LFR cluster, Figure 2C). Repeating the fastSTRUCTURE analyses for sites with recent stocking and the two hatcheries independently, UTA-2 was again the only individual above the stricter threshold ($q > 0.1$) with 20% membership to the LFR and LFA clusters combined (Figure S4A). There were 25 additional samples that had a combined 10%–20% membership to the LFR and LFA clusters (at $K = 9$), but in the independent analyses, all these had $q < 0.1$ (Figure S4).

3.3 | Effective Population Size and Migration Rates

Estimates of effective population size in wild populations varied between 22.3 (coastal site SBD) and 509.2 (UTA) with some sites showing large (infinite) 95% CIs. The estimates for the two hatcheries were among the lowest (17.8–33.6) from all sites with narrow 95% CI (Table 1). BayesAss indicated low to moderate levels of recent gene flow, both downstream and upstream, among sites within the same catchment. Despite poor convergence in some data sets, the posterior mean parameter estimates were consistent between replicated runs. Significant migration rates (estimated values with 95% CI not overlapping zero) generally occurred between sites with the lowest genetic differentiation (Figure 1B). For the first subset involving sites along the upper Wood River, there was support for unidirectional dispersal from BAK to both FRC and WR (Figure 1B). For the second subset in Beaver River, unidirectional gene flow from BRH upstream to BRO (0.11) was lower than downstream gene flow to BRP (0.22). A significant migration rate (0.05) was found from BRP to BHU but was not consistently significant between replicated runs. For the third subset in Meadow Brook, there was support for upstream dispersal from MBC to MBP (Figure 1B). For the fourth subset in Breakheart Pond tributaries, there was stronger support for downstream gene flow than upstream between UTB and UTB2 (Figure 1B).

3.4 | Riverscape Genetics

We used various approaches to explore the effects of riverscape features (i.e., habitat fragmentation and waterway distance) on population genetic diversity and gene flow among RI sites. There was a negative correlation between genetic diversity and increasing waterway distance from the downstream-most sampling site at UTA, albeit not significant (Figure 4: Pearson's correlation: $r_{\text{Pearson}} = -0.48$, $p = 0.080$; Figure S6). Also non-significant was the negative correlation between genetic diversity and the number of barriers from UTA (Figure 4: Pearson's correlation: $r_{\text{Pearson}} = -0.32$, $p = 0.27$; Figure S6). Sites below a dam had significantly higher genetic diversity than those above a dam but only for expected heterozygosity (Figure S3; Wilcoxon rank sum exact test, $p = 0.02$).

Multiple regressions and Mantel tests were used to evaluate three models explaining genetic differentiation among sites. The best fit model (lowest AICc) from the regression tests was M1 (isolation-by-resistance model) followed by M3, supporting a combined effect of waterway distance and number of dams on population differentiation (Table 2, Table S8). The ΔAICc was less than two, suggesting both models are equally distant from the truth, albeit M1 had a higher weight of evidence (Table 2). The Mantel tests were significant across all three models, revealing both an isolation-by-resistance (M1: $r_{\text{Mantel}} = 0.48$, $p = 0.0001$) and isolation-by-distance (M3: $r_{\text{Mantel}} = 0.4$, $p = 0.0035$) effect in the Rhode Island river networks (Table 2; Figure 4, Table S8).

3.5 | Body Shape Differentiation

The CVA permutation tests suggested that all seven groups were statistically distinguishable (p-value always < 0.001), but the analysis indicated that some groups were clearly more distinct in mean shape than others. The most obvious result of the CVA was that the first two canonical variates (CVs), which together accounted for 76.4% of the variation in shape between groups, effectively separated the trout into three groups:(1) the two hatchery strains, (2) the two coastal streams and (3) the five other inland streams (Figure 3A). Fish from both coastal streams tended to score low on CV1, and both groups tended to have large heads and short caudal peduncles (Figure 3A). Pairwise contrasts between trout groups from the DFA revealed that trout from the two coastal streams could be distinguished from each of the other groups with 98.5%–100% accuracy (Table S5). The fish from the two coastal streams were, however, similar in shape (Figure 3A). A pairwise comparison from the DFA indicated that fish from SBD and SHB were separated by small

TABLE 2 | Summary results from the regression and Mantel tests.

Model	F-statistic	Adj. R^2	AICc	ΔAICc	AICweight	r_{Mantel}	p_{Mantel}
M1: F_{ST} ~ Dams	$F_{1,89} = 26.67$	0.2219	−382.7	0.00	0.682	0.483	0.0001
M3: F_{ST} ~ Dams + Waterway distance	$F_{2,88} = 13.54$	0.2180	−381.0	1.62	0.303	0.294	0.0045
M2: F_{ST} ~ Waterway distance	$F_{1,89} = 17.36$	0.1538	−375.0	7.64	0.015	0.404	0.0035

Note: Three models were considered to explain genetic differentiation (F_{ST}); M1 as a function of the number of barriers (dams) between sites (isolation-by-resistance model), M2 as a function of waterway distance (isolation-by-distance model) and M3 testing the combined effect of waterway distance and dams. Models were evaluated and ranked based on ΔAICc. Mantel tests statistics (correlation 'r' and significance 'p') are based on 9999 permutations.

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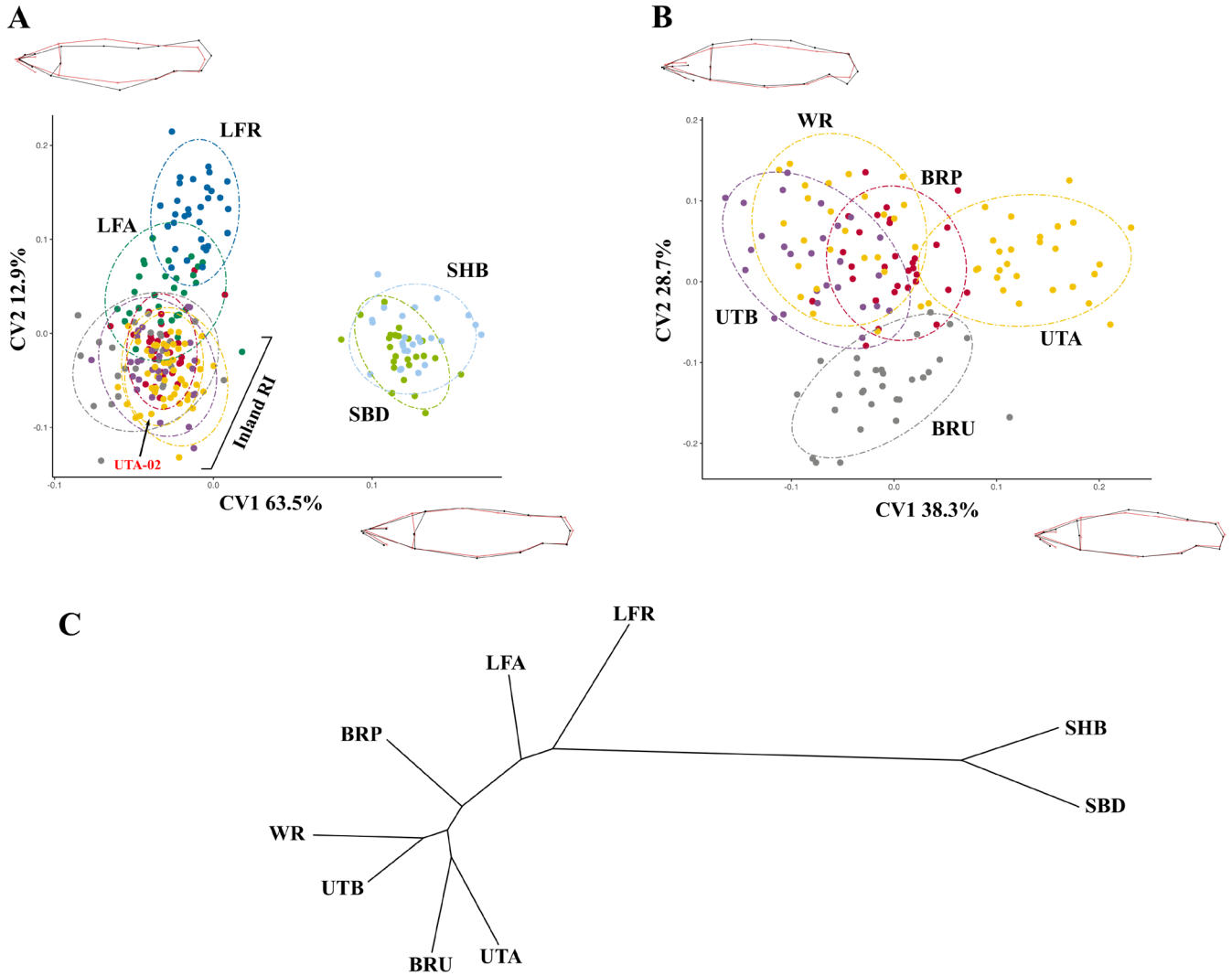

FIGURE 3 | Geometric-morphometric analysis of wild and hatchery-reared brook trout in Rhode Island. (A) Scatterplot of the first two canonical variates (CV1 and CV2), explaining 63.5% and 12.9% of the variation among all sites ($n = 9$) respectively with 90% equal-frequency ellipses highlighting the distinctiveness of the two coastal sites (SHB and SBD) and the two hatcheries (LFA, LFR) from the inland RI sites. The arrow shows the position of the single individual (UTA-02) that we found genetic evidence of hatchery introgression. (B) Scatterplot of the first two CVs which explain 38.3% and 28.7% of the variation respectively showing body shape morphological differences among the inland RI sites. The wireframe outline drawings both in (A) and (B) represent the positive extreme shapes of CV1 and positive extreme of CV2 (black outlines) compared to the consensus configuration (red outlines). Populations are colour-coded based on the fastSTRUCTURE results (at $K = 9$ and at $K = 6$, see also Figure 2C,D). (C) Visualisation of Mahalanobis distances among sites with a neighbour-joining tree.

Procrustes distances (0.014) and Mahalanobis distances (5.06) and classification accuracy was modest (73.5%). Fish from the two hatchery sites, LFR and, to a lesser extent, fish from LFA, generally scored high on CV2 and tended to have broad caudal peduncles, with short maxillae and small eyes set close to the top of their heads (Figure 3A). The pairwise contrast between the two hatchery groups illuminated the more subtle differences in body shape between them. Fish from LFA and LFR were separated by relatively large Procrustes (0.058) distance, and the two groups could be classified accurately (88.5%, Table S5). Fish from the LFA group (Assinica Strain) tended to have larger heads and slightly deeper bodies than fish from the LFR group (Rome Strain, Figure 3A).

Differences in mean body shape between the five inland stream groups were small relative to differences between these groups and the two hatcheries and the two coastal streams. Pairwise comparisons from the DFA showed that the five inland streams were separated by relatively small Procrustes distances (0.019–0.03) and Mahalanobis distances (3.98–6.05) and classification accuracy was generally lower for inland stream—stream pairs (67.3%–88.3%) than for inland stream—hatchery pairs (88.2%–96.4%) and coastal stream—inland stream pairs (98.1%–100%, Table S5). A second CVA limited to just these five streams indicated that all five groups were statistically distinguishable (permutation tests, p-value always < 0.001). CV1 and CV2 explained 67% of the variation in shape between these groups and showed that the two non-stocked groups (UTB and WR) were the most similar in body shape, whereas trout from the three stocked streams were slightly more distinct in body shape (Figure 3C). Pairwise comparisons from the DFA showed that trout from UTB and WR were separated by the smallest

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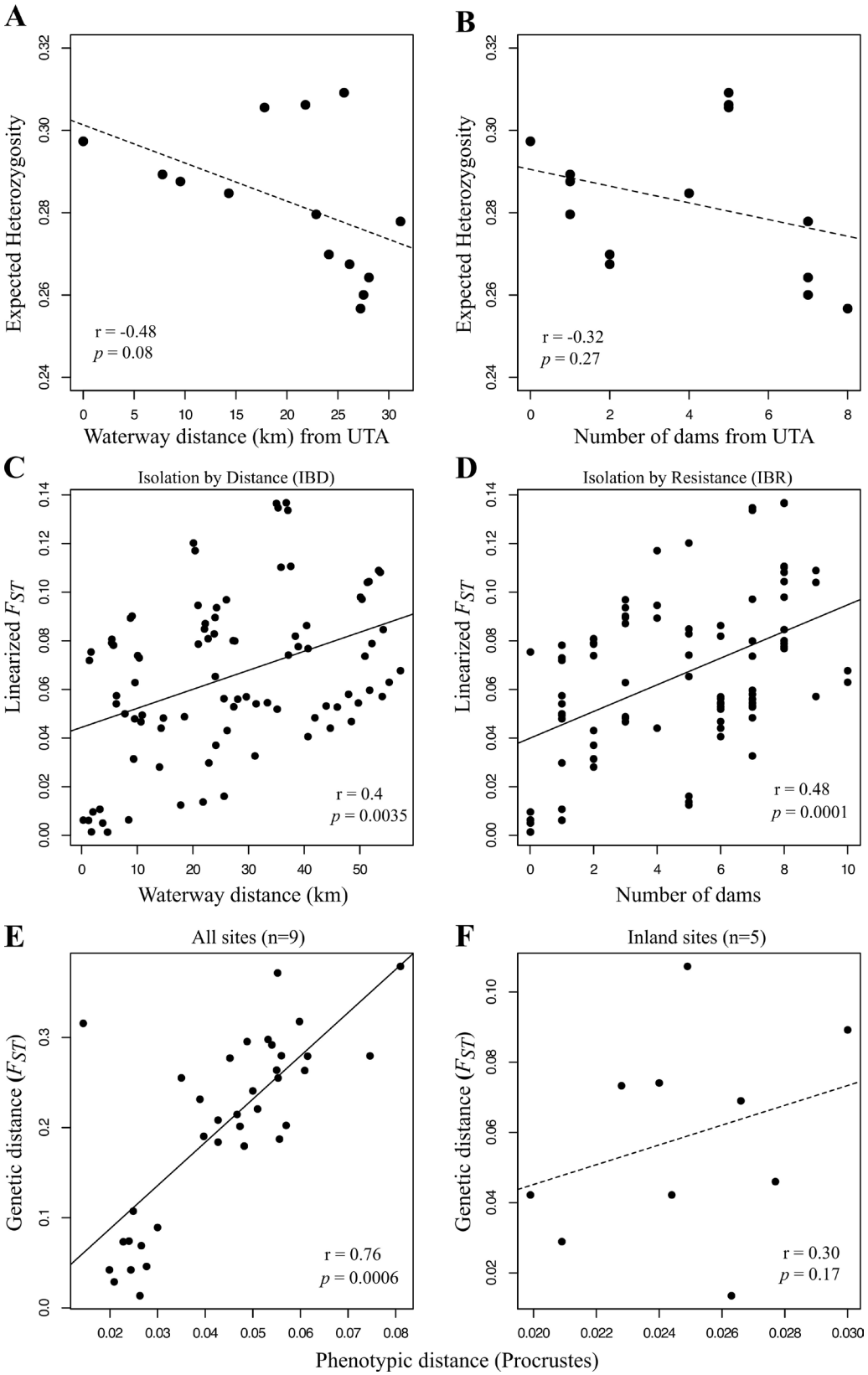

FIGURE 4 | Relationships between genetic diversity (expected heterozygosity), genetic differentiation (F_{ST}), riverscape features (waterway distance, dams) and phenotypic differentiation (P_{ST}). Negative correlation between expected heterozygosity and (A) waterway distance and (B) number of dams from the downstream-most site at UTA. (C) Significant association between linearized F_{ST} and waterway distance testing for an isolation-by-distance (IBD) model between all wild sites. (D) Significant association between the number of dams and linearized F_{ST} testing for an isolation-by-resistance model (IBR). (E) Significant association between genetic and phenotypic (Procrustes) distances of all populations used in the morphometric analyses and (F) non-significant association between genetic and phenotypic (Procrustes) distances of wild sites only. Correlation statistics from Mantel tests (*r* and significance) are shown within each panel, with non-significant associations indicated with a dashed line.

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Procrustes (0.022) and Mahalanobis distances (3.98, Table S4) and classification accuracy (67.3%) was not much better than random (50%, Table S5).

There was a significant association between phenotypic distance (Procrustes) and genetic differentiation (F_{ST}) between inland and coastal sites and the two hatcheries (Figure 4C: $r_{Mantel} = 0.76$, $p = 0.0006$). Within inland sites, the association was positive but not significant (Figure 4D: $r_{Mantel} = 0.30$, $p = 0.16$).

4 | Discussion

In this study we developed over 16K SNP markers to assess population genetic structure and the impact of anthropogenic activities on wild brook trout in RI rivers. Overall levels of genetic diversity were low, but higher in most wild populations than the two captive-bred strains. We found moderate population genetic structure within RI, with streams belonging to the same catchment being genetically more similar and grouped together in all the clustering analyses. Our results showed minimal anthropogenic impact on wild brook trout via stocking, but there was an effect of habitat fragmentation due to dams. Evidence for hybridisation between captive-bred and wild fish was very low and we found greater support for an isolation-by-resistance model. We also found both uni- and bi-directional as well as upstream and downstream gene flow among pairs of sites within the same catchment despite the presence of dams. Phenotypic differentiation also was apparent in our system both between the wild and the hatchery individuals and within RI streams. Specifically, the two coastal sites, SBD and SHB, differ substantially from the inland RI sites both in their genetic diversity and structure and morphology (body shape), reflecting long-term isolation in the absence of gene flow and possibly local adaptation. These results highlight the importance of targeting intraspecific variation in conservation planning for vulnerable populations.

Species important for subsistence, recreational and commercial fisheries have experienced major population declines as a result of cumulative impacts. Every year, thousands of hatchery-reared fish are released into the wild to compensate for reduced yields and bolster abundance and to support the recreational activity of anglers. Such practice, however, could be associated with a range of negative effects, from ecological (e.g., intraspecific competition for food and resources, Einum and Fleming 2001) to evolutionary (e.g., decreases in genetic diversity and loss of local adaptation due to interbreeding and transfer of maladaptive genes, Fraser 2008). In our study region, we found minimal evidence (one individual, 0.3% of wild fish) for introgression between wild and captive-bred brook trout. This result was also supported by the strong genetic (F_{ST} range: 0.17–0.38) and phenotypic divergence (Procrustes distance range: 0.035–0.081) between the two hatchery strains and the wild populations, suggesting that interbreeding was rare. However, we found higher levels of genetic diversity in stocked sites compared to non-stocked sites. This could be due to the differential effects of genetic drift (stocked sites being relatively more connected) and/or streams with characteristics that may support larger populations (Neville et al. 2006; Whiteley et al. 2013), rather than the introduction of new alleles through hatchery introgression. The low rates of introgression we observed in our study could be explained by the poor survival of hatchery individuals in the wild because of artificial selection (Araki et al. 2008; Webster and Flick 1981) along with high angling mortality. If multiple hatchery strains are used to stock brook trout populations, however, it is possible that one type is successful at surviving long enough to interbreed (Bruce, Kutsumi, et al. 2020). Stocking pressure has been positively correlated with introgression in salmonid populations (brown trout, Hansen and Mensberg 2009; brook trout, Marie et al. 2010; lake trout, Valiquette et al. 2014). Other studies that have found extensive introgression in wild brook trout populations have also identified variables associated with introgression that include watershed area and water chemistry (Bruce, Kutsumi, et al. 2020) and sulfates (Bruce, George, et al. 2020). Alternatively, other studies have demonstrated that hybridization between captive-bred and wild fish may not occur at all or may occur only at low levels (Bruce and Wright 2018; Lehnert et al. 2020; Létourneau et al. 2018; White et al. 2018). The variation in the outcomes of stocking could also be in part due to the genetic tools used (i.e., low number of microsatellite markers), the year since the mean year of stocking (Létourneau et al. 2018), but also in the details of the hatchery strains; local vs. non-local stock and their overall fitness (Araki et al. 2008). The origin of the strains could play a major factor if these originated from a different part of the brook trout's native range (e.g., northern inland lakes and rivers). Supplemental stocking has been occurring for decades in the RI region, with the two hatchery strains (LFA and LFR) accounting for all recorded, recent (< 10 years) stocking events in the study streams. However, comprehensive historical stocking data are not readily available, and we cannot exclude the possibility of other occasional undocumented stocking events (e.g., fish from an unknown strain) over the past 100 years. Thus, we might have underestimated the level of hatchery introgression in this system.

One of the main causes of fragmentation of freshwater habitats has been the widespread construction of dams for hydroelectric power plants, drinking water retention, watermills, and recreational activities (Barbarossa et al. 2020). Their impacts on freshwater ecosystems are well recognised (e.g., Zarri et al. 2022) and range from modifications to population genetic diversity (brown trout, Horreo et al. 2011) to changes in species assemblages (Turgeon et al. 2019). In Rhode Island, the spatial structure of genetic diversity and population divergence suggests a moderate influence of habitat fragmentation. There was a downstream increase in genetic diversity (DIGD), but it was not significant nor correlated with the number of barriers from the downstream-most site. The presence of dams may have partially disrupted the expected natural downstream-biased gene flow (e.g., Horreo et al. 2011) but other processes (e.g., colonisation history and habitat variability) can generate DIGD even in unfragmented habitats (Paz-Vinas et al. 2015). However, when considering population differentiation, we found greater support for the isolation-by-resistance model, with limited gene flow occurring mainly among pairs of sites within the same catchment (even in the presence of dams). The BayesAss analysis, though, is sensitive to population genetic structure, and the low levels of genetic differentiation observed among our sampling sites might have overestimated the calculations for a recent immigration (Meirmans 2014), which might better reflect shared ancestry

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before the construction of the dams and habitat fragmentation. Other studies on brook trout that have looked at genetic structure in dendritic stream networks have suggested that the species in these systems exhibits relatively little movement (Hudy et al. 2010; Kanno et al. 2011) and that waterfalls disrupt their gene flow (Kelson et al. 2015; Torterotot et al. 2014). The effects of human-mediated fragmentation on the genetic structure of wild populations might be difficult to disentangle from historical evolutionary processes (Epps and Keyghobadi 2015) and might be different in each study system. Indeed, by examining five fish species with varying life histories above and below a dam in the lower Boardman River, Michigan, USA, Gehri et al. (2021) found heterogeneous patterns of genetic structure within each species highlighting the complex dynamics of fragmented river systems. Nevertheless, our results do show an influence of dams on gene flow, but it should be noted that these are not absolute barriers. Artificial structures such as dams, weirs, and culverts impose different levels of movement restrictions for aquatic organisms that can have directional bias (downstream movement is often easier than upstream movement) and depend on water levels (Thomaz et al. 2007).

Morphological variation in trout populations can reflect both adaptive and nonadaptive genetic divergence among populations plus phenotypic plasticity in response to the environment (e.g., Gossieaux et al. 2020; Keeley et al. 2021; Pakkasmaa and Piironen 2001). The isolation of animals can lead populations to genetically diverge, which can be expressed as phenotypic differences. Brook trout are known to display variable morphologies, and we found strong phenotypic differences between wild and hatchery brook trout that also correlated with genetic divergence. This result was expected, as it also has been shown in other studies on brook trout (e.g., Gossieaux et al. 2020). The distinct morphological difference between wild and hatchery brook trout can have management implications in that morphological analysis of brook trout photographs could be used to detect hatchery-raised individuals or their hybrids caught in the wild. McKinney et al. (2014) compared the genetic and morphological divergence between three strains of brook trout including the Assinica strain used for stocking in RI, which they found to have diverged more than the other two possibly due to differences in artificial selection in hatcheries. Our analysis, however, showed that the Assinica strain (LFA) is more similar in morphology (body shape) to the inland RI stream brook trout than the other strain (LFR), but both hatchery strains are genetically very distinct. We also found a significant association between phenotypic and genetic distance, which was mainly due to the strong divergence between the hatchery and the coastal sites compared to the remaining inland RI stream populations. Within RI, the association was positive but not significant despite the apparent morphological differences between some population pairs. A lack of association may reflect small sample sizes and subtle morphological differences but also points to plastic responses to habitat heterogeneity as shown in Swiss populations of brown trout (Stelkens et al. 2012). In fragmented brook trout populations, natural and sexual-selective pressures on phenotypic traits might be stronger and consistent with habitat conditions (Zastavniouk et al. 2017). Morphometric analysis has become a widely used tool for population delineation (Cadrin 2000) and, because it is more time- and cost-efficient than the genetic analyses, our results support its future use to pre-screen brook trout populations for potential genetic divergence.

Restricted gene flow and genetic drift likely explain the genetic and phenotypic differentiation we observed in the two coastal populations in RI. However, like other salmonids, brook trout can be anadromous wherever there is free access to the sea and marine or freshwater habitats remain sufficiently cool throughout the summer (Dauwalter et al. 2014). The current status of the two coastal populations in Rhode Island with regard to anadromy is unknown. Historically, anadromous brook trout occurred along the coast of southern New England, although their number has decreased due to habitat alteration and overfishing (Dauwalter et al. 2014). In Cape Cod, Massachusetts, wild-reproducing coastal populations occur in a few remnant tributaries (Annett et al. 2012), but the degree to which the individuals use the ocean is unclear. These coastal brook trout streams also were heavily stocked between 1940 and 1980s, but with little evidence of introgression (Annett et al. 2012). Further sampling along the coast of RI to identify other isolated sites would enhance our understanding of the connectivity of these populations in southern New England and whether they represent anadromous stocks of the species that warrant separate management.

Levels of genetic diversity and effective population sizes were generally low but higher in wild populations than in the two hatchery strains. With some evidence that dams restrict gene flow, whether these wild populations can maintain the low levels of genetic diversity in the face of novel pressure (e.g., climate change) is unknown. Restoration management to increase connectivity between sites might be necessary (e.g., removal of dams) and additional sampling, to include more locations above and below dams, coupled with more complex riverscape genetic models (e.g., Nathan et al. 2019) would be needed to better assess the genetic consequences of restricted gene flow. With regards to ecological factors, analysing habitat variability and quality is essential to identify locally adapted populations. The two coastal populations might be locally adapted to the environmental conditions (e.g., temperature, salinity tolerance) and future conservation efforts should focus on maintaining their genetic diversity, critical for adaptive response to climate changes. Our outlier analyses showed 1070 loci that could be potentially under selection. A natural next step would be to analyse these candidate adaptive SNPs and test for associations with environmental variables and/or anadromy (e.g., Ferchaud et al. 2020). Understanding the origins and architecture of intraspecific variation is paramount for predicting the short- and long-term responses of species to novel and changing environments (Mimura et al. 2017). Our results will help inform management decisions to conserve wild brook trout in RI while adding valuable knowledge to the growing list of studies assessing the impact of anthropogenic activities across the native range of this species.

Author Contributions

Conceptualisation: T.M., A.L., C.P., G.F. and S.N.M. Developing methods and conducting the research: S.N.M., A.L., C.P., G.F. and T.M. Data analysis: S.N.M., C.J.F. and G.F. Data interpretation: S.N.M., T.M., C.P.,

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A.L. and G.F. Preparation of figures and tables: S.N.M. Writing: S.N.M., T.M., G.F., C.P., A.L. and C.J.F.

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Disclosure

Benefit-Sharing Statement: A research collaboration was established with the Rhode Island Department of Environmental Management (RIDEM) Division of Fish and Wildlife to co-develop the project, coordinate fieldwork, and provide samples, with the main collaborators included as co-authors. The results of this research have been shared with the RIDEM, the general audience, and the broader scientific community. This project addresses a priority concern, in this case brook trout, as a species of Greatest Conservation Need in the region. Additional benefits from this research occur from the sharing of data and results on public databases as described above.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Genetic data of the full dataset (BT_All; 444 samples, 16,336 SNPs; in vcf format) as well as data from the geometric-morphometric analysis (fish fork length and canonical variate scores) are available on *figshare* DOI: https://doi.org/10.6084/m9.figshare.28628804.v1. Additional data can be found in Tables S1–S8 and Figures S1–S6; and include (a) pairwise F_{ST} (Table S3), (b) Mahalanobis and Procrustes matrices (Table S4), (c) number of dams between sites (Table S5), and (d) waterway distance between inland sites (Table S7).

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